

Image Enhancement in the Spatial Domain

Histogram Processing

Image Histogram

- Image Histogram
 - For describing a digital image with gray levels $[0, L-1]$
 - $p(r_k) = n_k/N$ is the probability of occurrence of gray level r_k
 - r_k is the k^{th} gray level
 - n_k = number of pixels with k^{th} gray level
 - N = total number of pixels
 - $k = 0, 1, 2, 3, 4, 5, \dots, L-1$

Image Histogram

- $p(r_k)$ is the probability of the occurrence of gray-level r_k
- $p_r(r_k)$ is the probability density function (PDF) of the variable $r_k, k = 0, \dots, L - 1$
- Here, $p_r(r_k)$ and $p(r_k)$ are used interchangeably

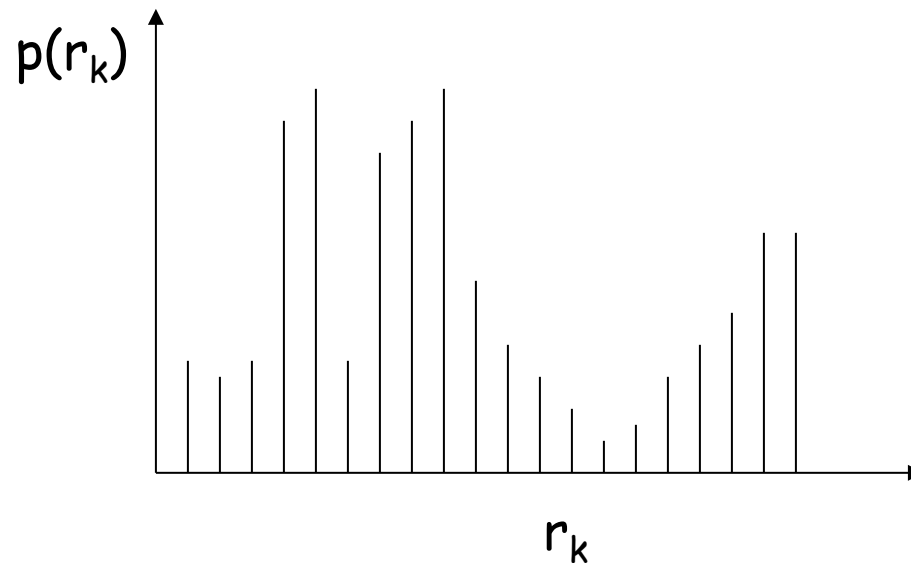


Image Histogram

- A histogram can tell you a lot about an image
- Gray level intensity distribution
- It can be modeled by a statistical distribution (red line and continuous)

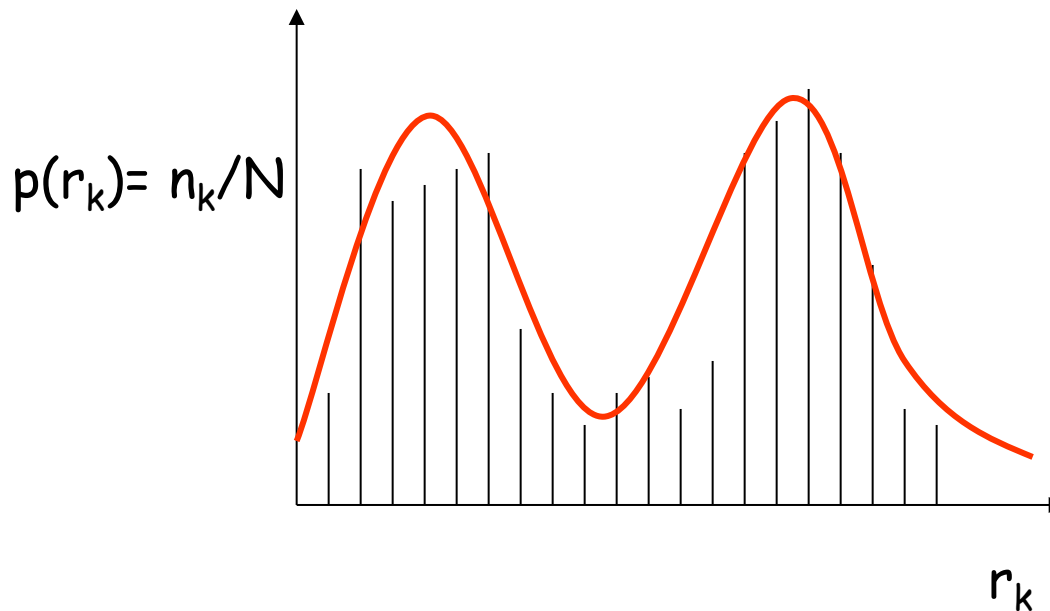


Image Histogram

- Gray level intensity distribution helps define an intensity threshold

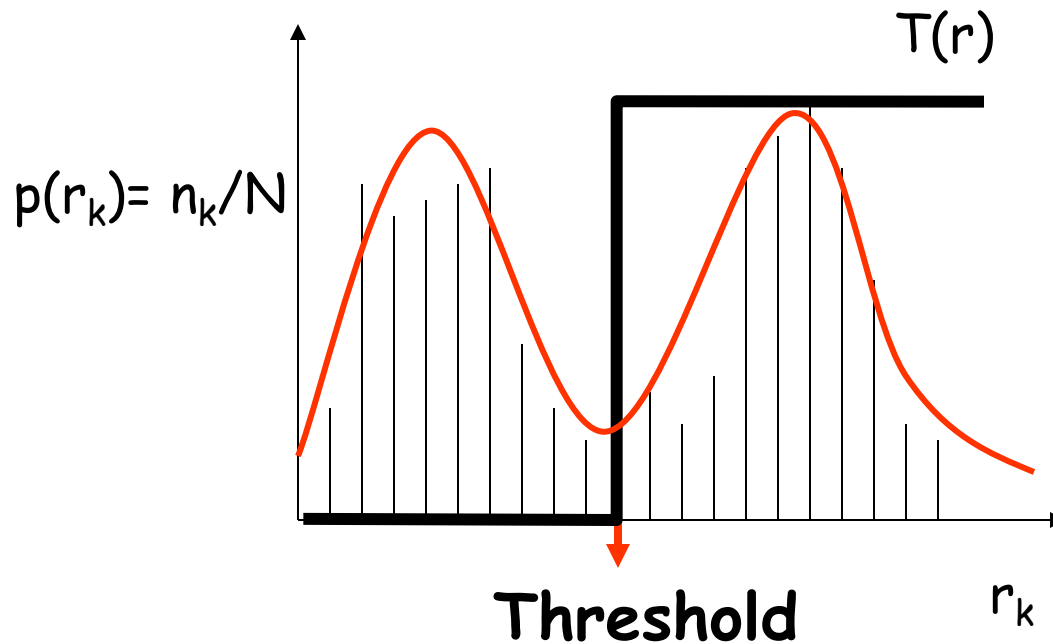


Image Histogram

- The two images below give the same image histogram, and this histogram cannot differentiate the two images.
- No spatial information, e.g., relationship between pixels, is captured in the histogram.

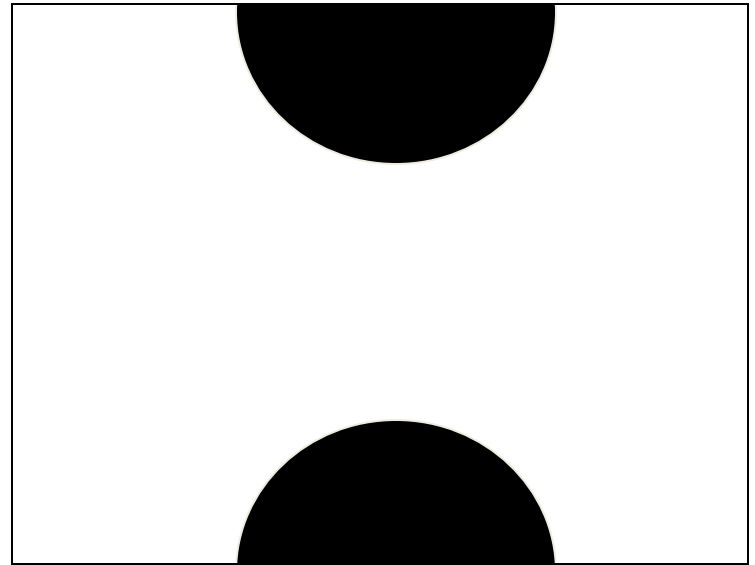
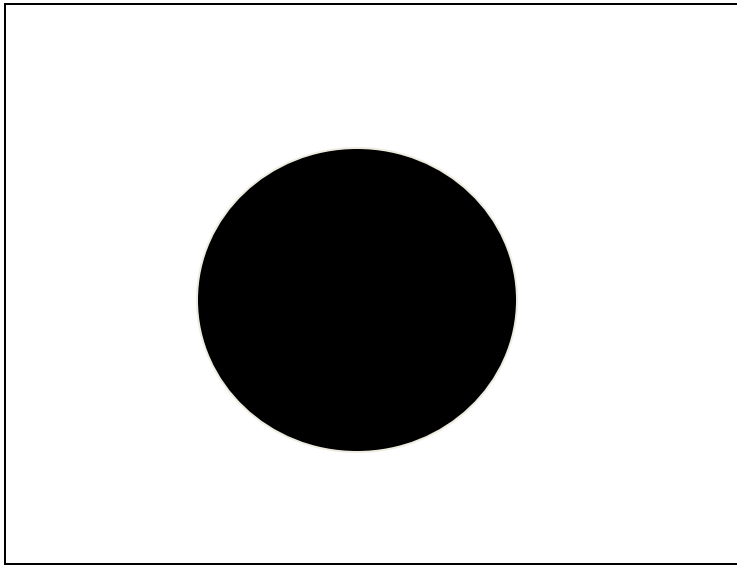
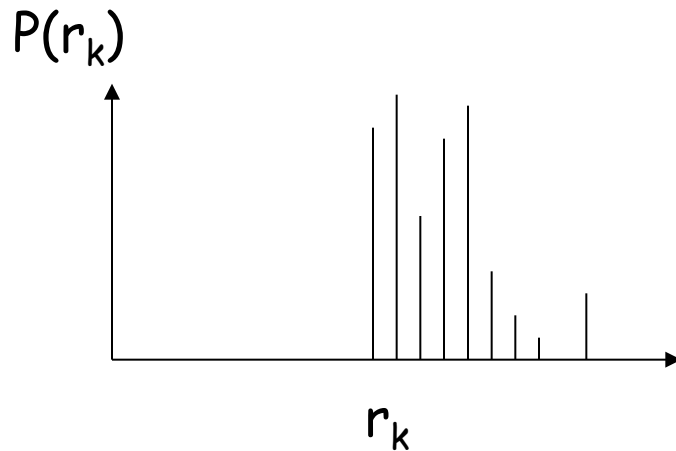


Image Histogram

- Image types

Bright Image



Dark Image

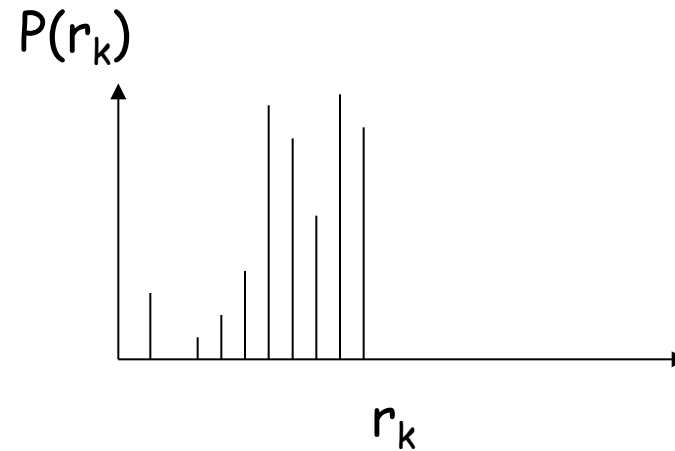
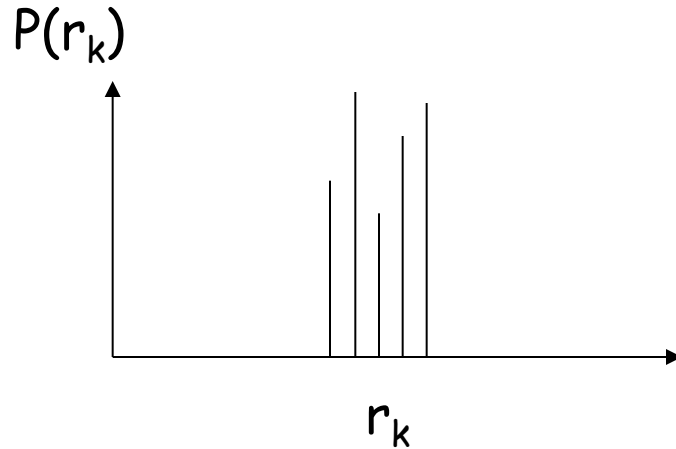


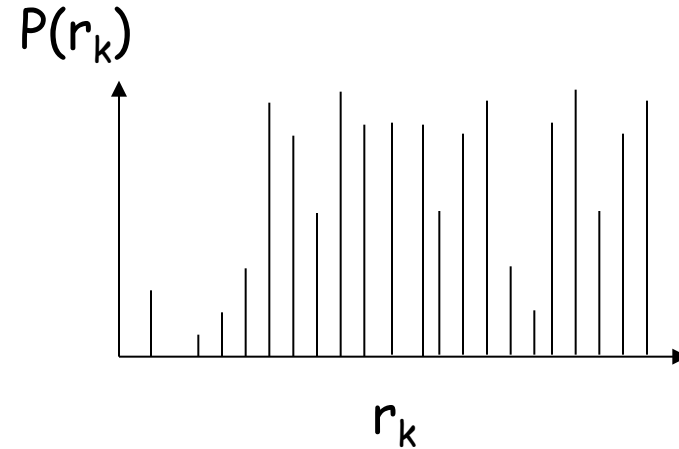
Image Histogram

- Image types

Low Contrast



High Contrast



Some Examples

For a,b,c, narrow intensity ranges have poor contrast

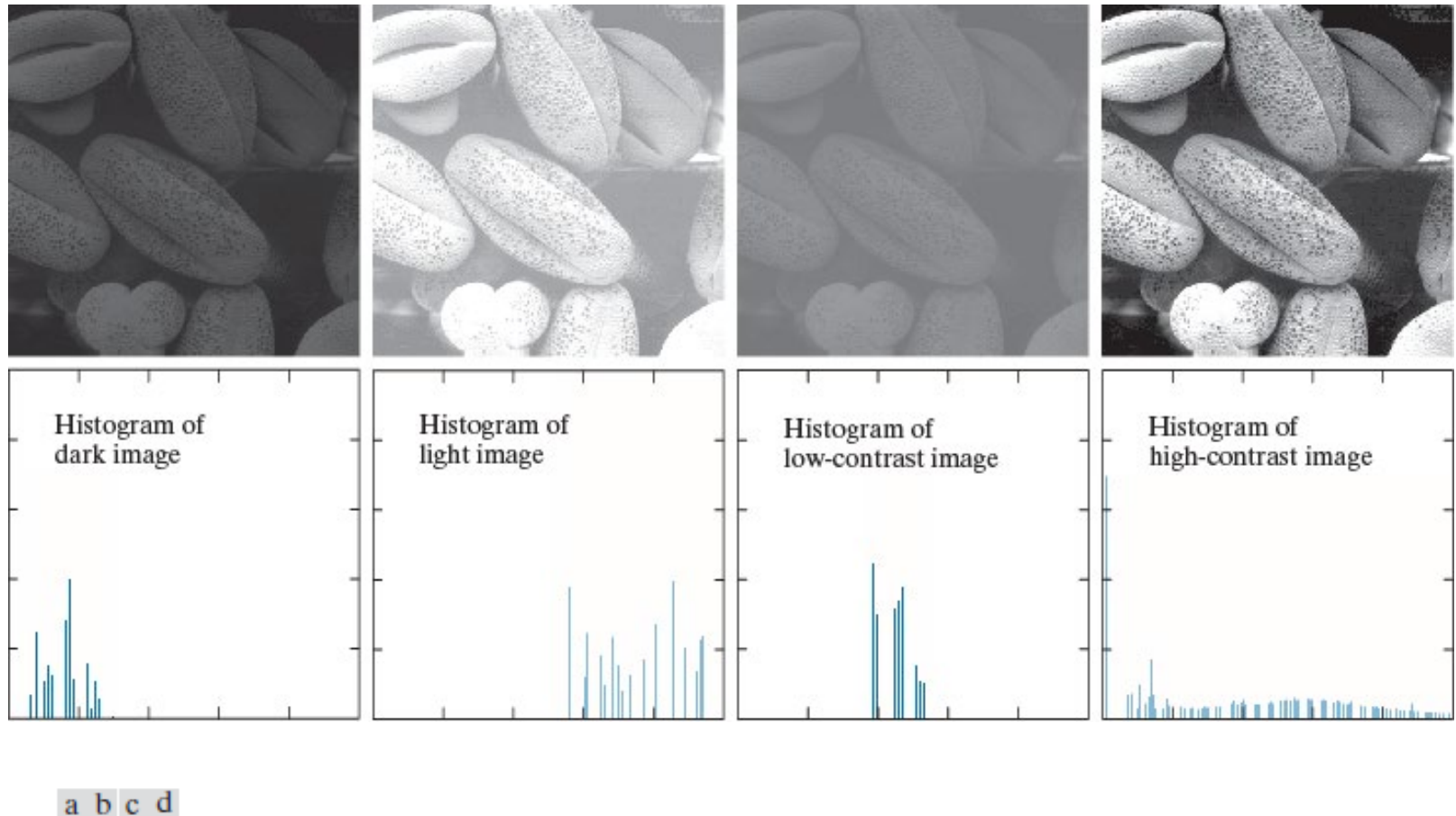
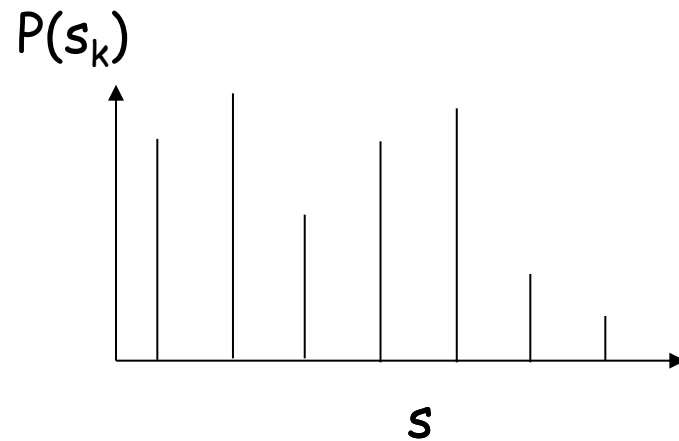
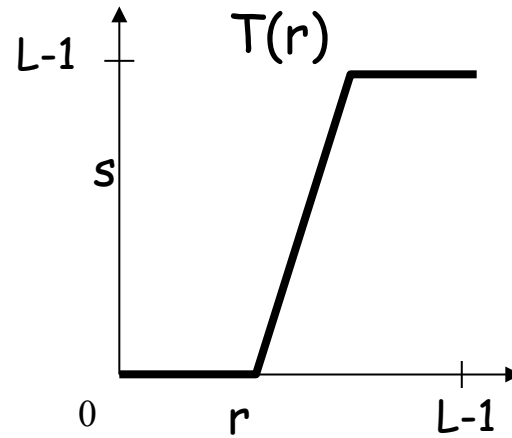
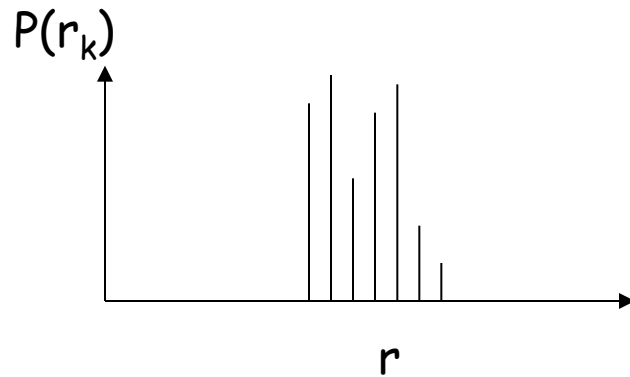


FIGURE 3.16 Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of r_k and the vertical axis are values of $p(r_k)$.

Contrast Stretching

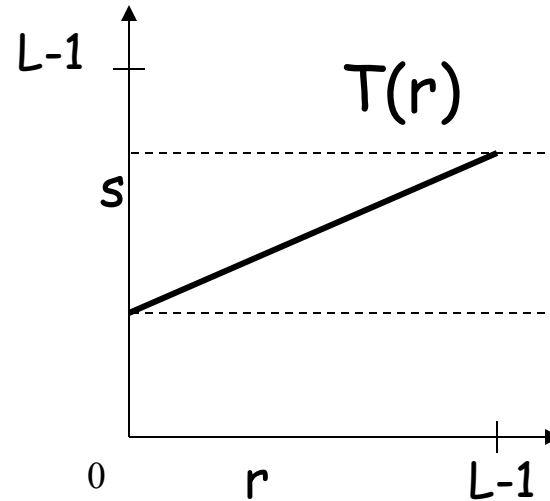
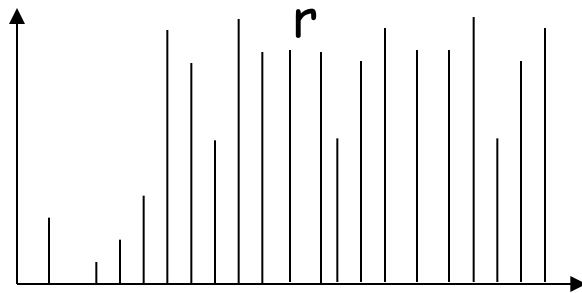
Low Contrast



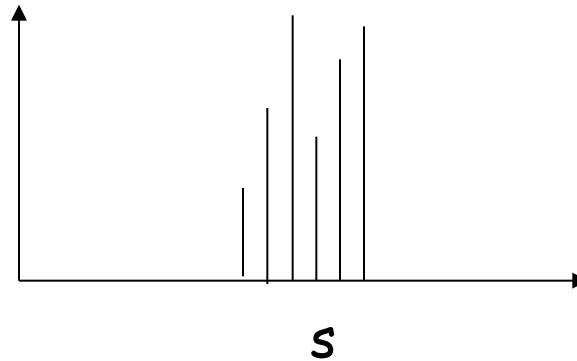
Contrast Compressing

High Contrast

$P(r_k)$



$P(s_k)$



Histogram Equalization

- We want an image with equally many pixels at every gray level, or the output intensity approximately follows a *uniform distribution*.
- That is, a flat histogram, where each gray level, r_k , appears (N/r_m) times
 - where “ r_m ” is the maximum gray level
 - N is the number of pixels in the image
- Example: <https://demonstrations.wolfram.com/HistogramEqualization/>
- The goal is to spread out pixel values across the full intensity range so that the low contrast intensity ranges gain high contrast in the output.

Histogram Equalization

- The weighted cumulative distribution function (CDF) of r is represented by (in continuous form), (T is the intensity transformation function)

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

- This transformation function seeks to generate an output image with a uniform (flat) probability density function (PDF), independently of the form of $p_r(r)$

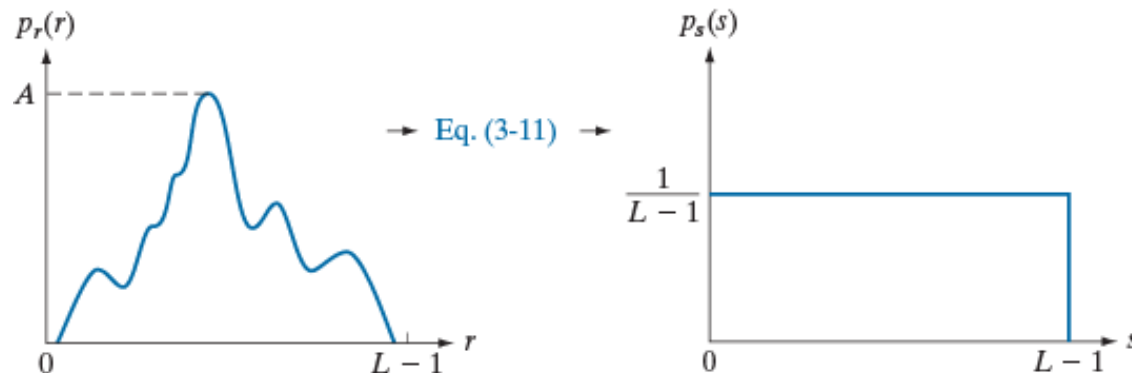


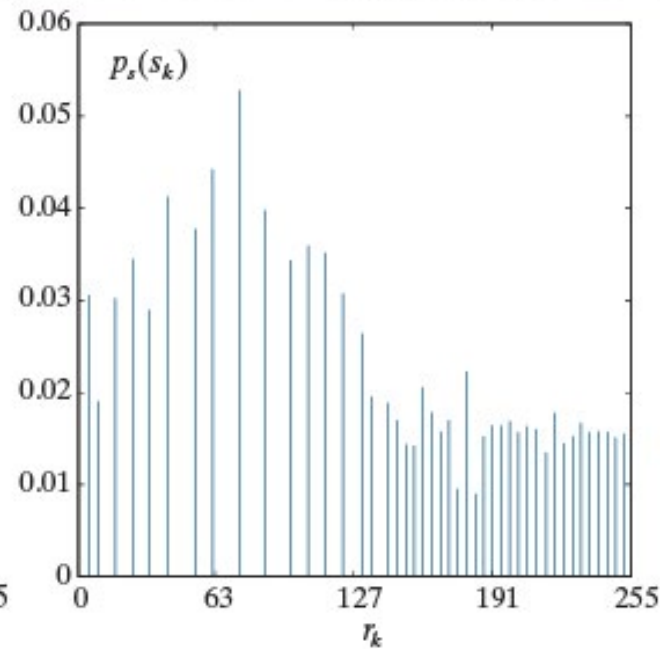
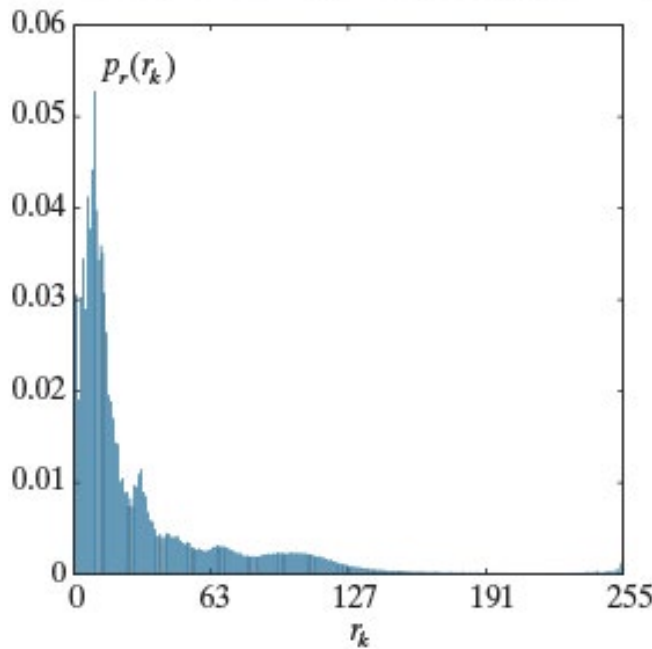
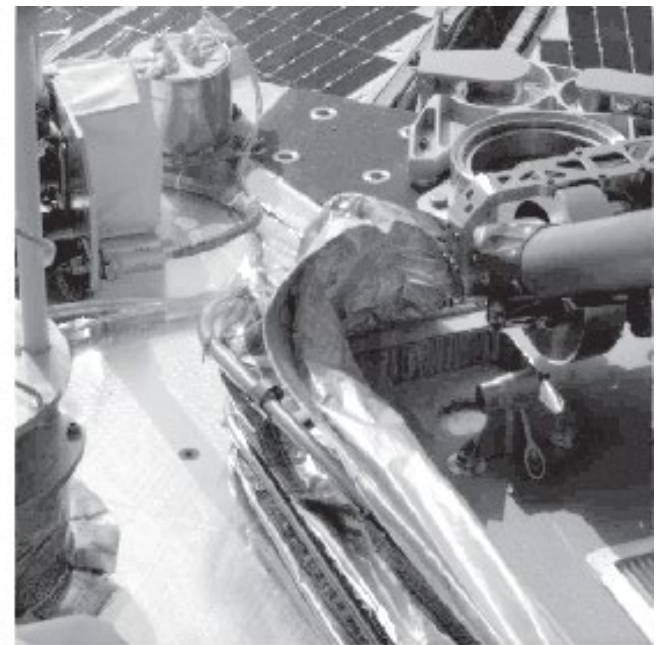
FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying Eq. (3-11) to the input PDF. The resulting PDF is always uniform, independently of the shape of the input.

Why does histogram equalization give a uniform distribution?

a b
c d

FIGURE 3.22

(a) Image from Phoenix Lander.
(b) Result of histogram equalization.
(c) Histogram of image (a).
(d) Histogram of image (b).
(Original image courtesy of NASA.)



Use $T(r)$ to equalize the histogram

- Assume the input variable r has been normalized between $[0,1]$
- $s = T(r)$, there are two properties
 - (a) $T(r)$ is single-valued in the interval $0 \leq r \leq 1$
 - (b) $0 \leq T(r) \leq 1$ for $0 \leq r \leq 1$

Conditions (a) and (b) for implementation

- (a) $T(r)$ is single-valued and non-decreasing in the interval $0 \leq r \leq 1$
 - Can preserve the order from black to white in the gray scale.
 - Can preserve the basic appearance of an image.
 - If T is strictly monotonic and increasing, it is invertible.
- (b) for $0 \leq r \leq 1$, $0 \leq T(r) \leq 1$
 - Can guarantee a mapping that is consistent with the allowed range of pixel values.
 - No intensity rescaling is needed.

Discrete approximation of T

- Let's look at the discrete approximation

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k \frac{n_j}{N} = (L - 1) \sum_{j=0}^k p_r(r_j)$$

s_k is the output intensity value, $0 \leq s_k \leq L-1$

r_k is the input intensity value, $0 \leq r_k \leq L-1$

n_j is the number of pixels with the j^{th} gray level

$k = 0, 1, 2, 3, \dots, L-1$ (gray levels)

Example

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

TABLE 3.1
Intensity
distribution and
histogram values
for a 3-bit,
 64×64 digital
image.

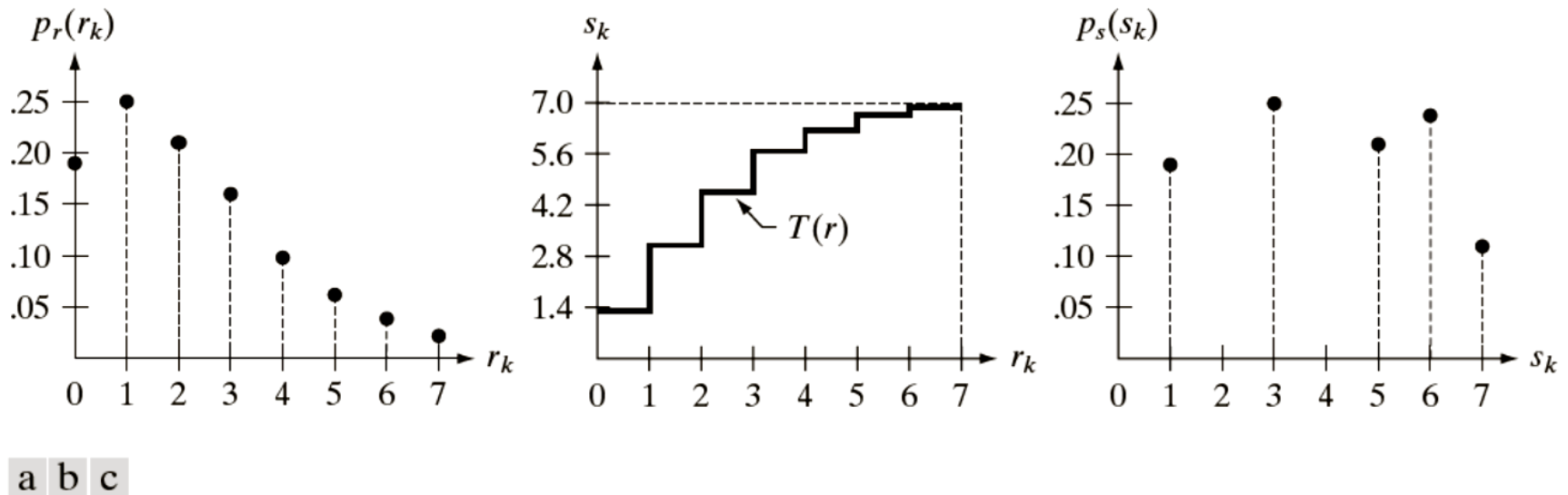
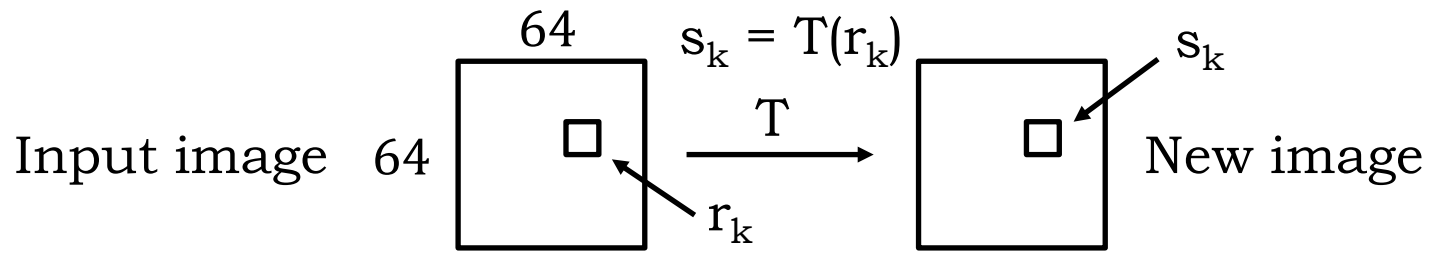
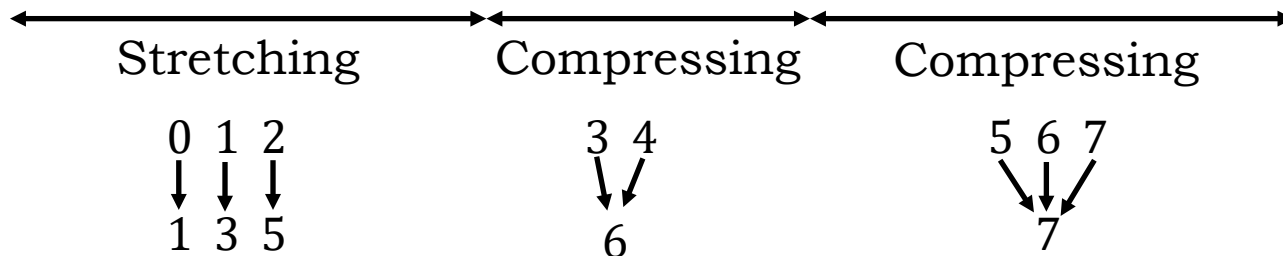


FIGURE 3.19 Illustration of histogram equalization of a 3-bit (8 intensity levels) image. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.



Total number of pixels = $64 \times 64 = 4096$ pixels

k	0	1	2	3	4	5	6	7	$\frac{2^3-1=7}{2^8-1=255}$
r_k	0	1	2	3	4	5	6	7	
$Pr(r_k)$	$\frac{790}{4096}$ 0.19	$\frac{1023}{4096}$ 0.25	$\frac{850}{4096}$ 0.21	$\frac{656}{4096}$ 0.16	$\frac{329}{4096}$ 0.08	$\frac{245}{4096}$ 0.06	$\frac{122}{4096}$ 0.03	$\frac{81}{4096}$ 0.02	$\sum Pr(r_k) = 1$
$S_k^{*(1/7)}$	0.19	0.44	0.65	0.81	0.89	0.95	0.98	1	
Gray levels	1	3	5	6	6	7	7	7	



The discrete formulation of *histogram equalization* is given by

$$s_k = T(r_k) = \sum_{j=0}^k p_r(r_j), \quad k = 0, 1, 2, \dots, L-1$$

where r_k and s_k represent the original and transformed grey levels respectively, L denotes the number of discrete grey levels, T is the transformation function, p_r is the probability density function

$p_r(r_j) = n_j/n$, n is the total number of pixels and n_j is the number of pixels having intensity level r_j .

(a) Perform histogram equalization on a 3×3 input image, as shown below. Show all steps.

Answers:

Input image

0	0.33	0.67
0.33	1	0.33
0.67	0.33	0.67

Histogram equalized image

0.11	0.56	0.89
0.56	1	0.56
0.89	0.56	0.89

Round to 2 decimal places

	r_k	0	0.33	0.67	1
	$P_r(r_k)$	$1/9$	$4/9$	$3/9$	$1/9$
Transformation function	S_k	$1/9$	$5/9$	$8/9$	$9/9$
		$= 0.11$	$= 0.56$	$= 0.89$	$= 1$

(b) Explain why the transformation function $T(r_j)$ is increasing in the interval $0 \leq r_j \leq 1$, where $j = 0, 1, 2, \dots, L-1$.

Answers:

$$S_k = T(r_k) = \sum_{j=0}^k \Pr(r_j)$$

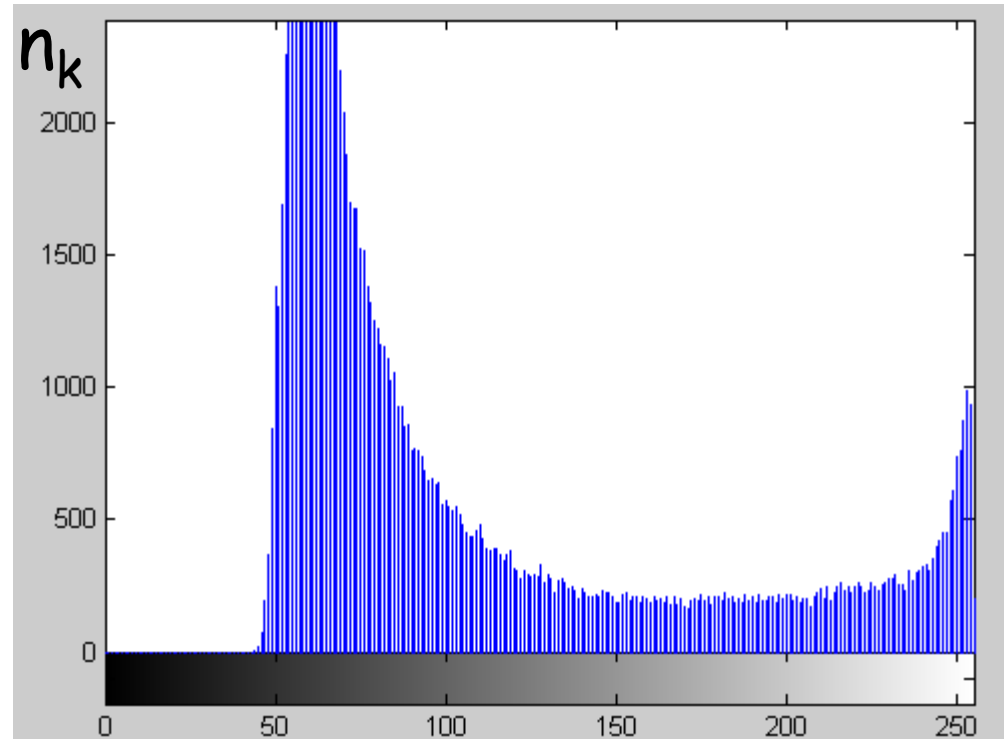
$\Pr(r_j)$ is non-negative

Example

<http://www.mathworks.com/help/toolbox/images/ref/histeq.html>



Image

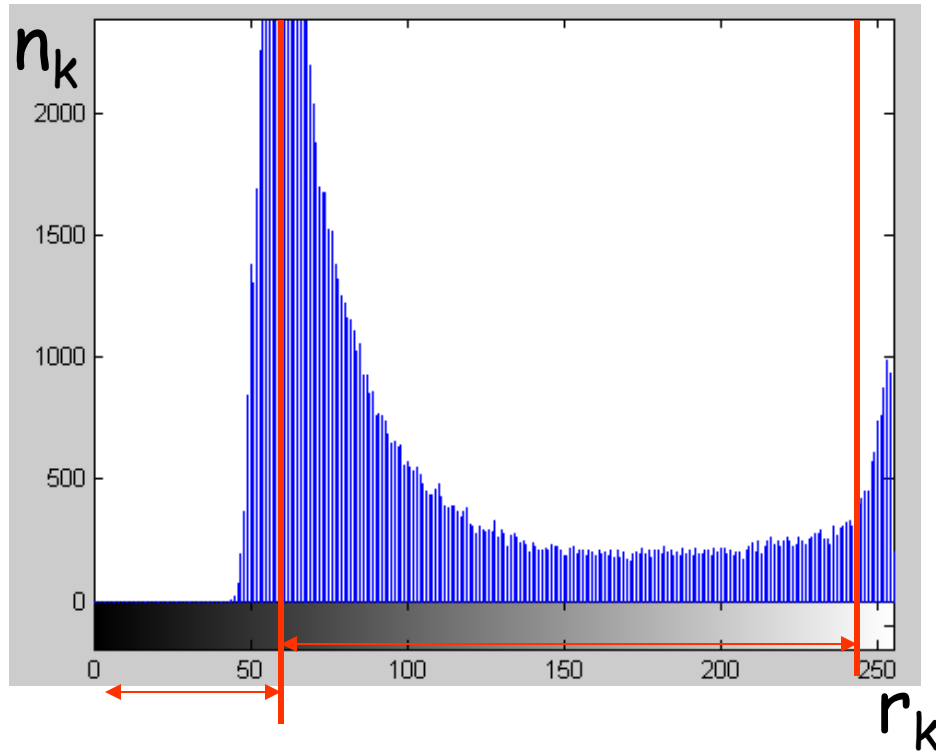


Histogram

Notice, this is not normalized, y axis is n_k .

To normalize, let y axis = $p(r_k) = n_k / N$

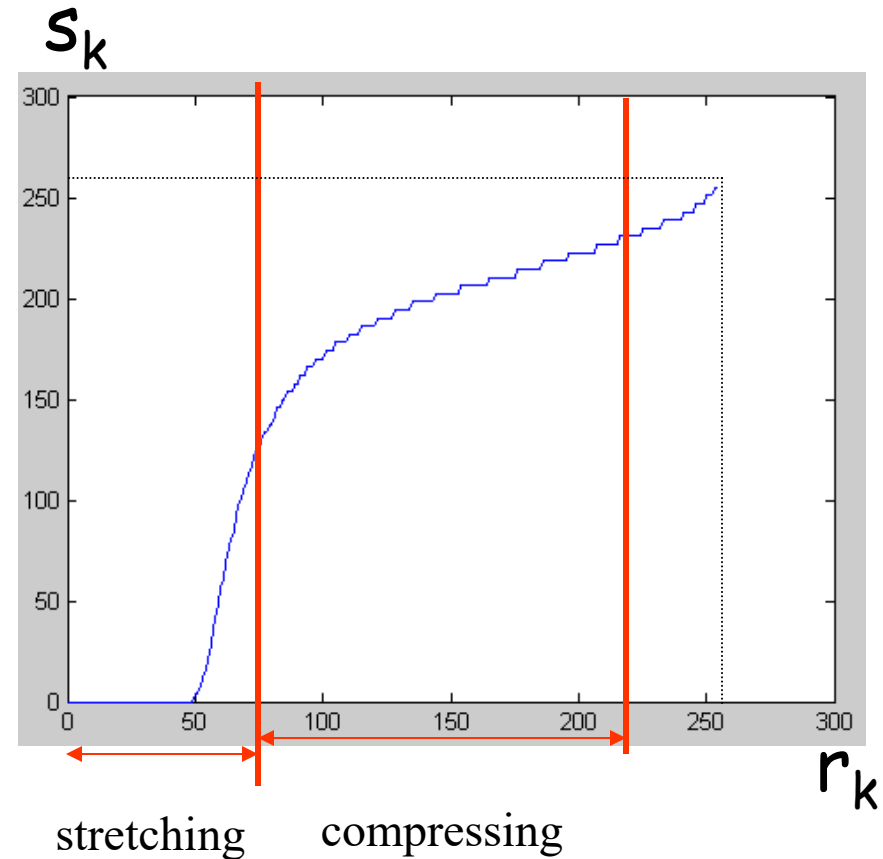
Example



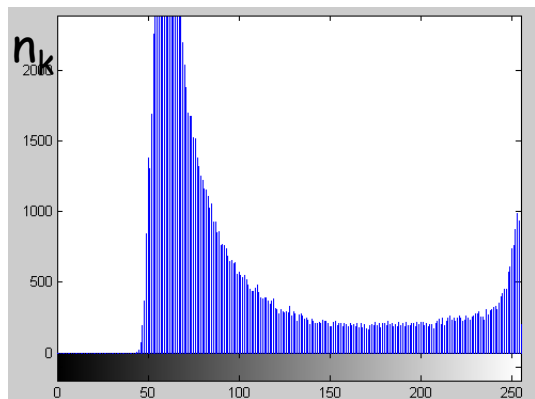
Histogram

Notice, this is not normalized, y axis is n_k .

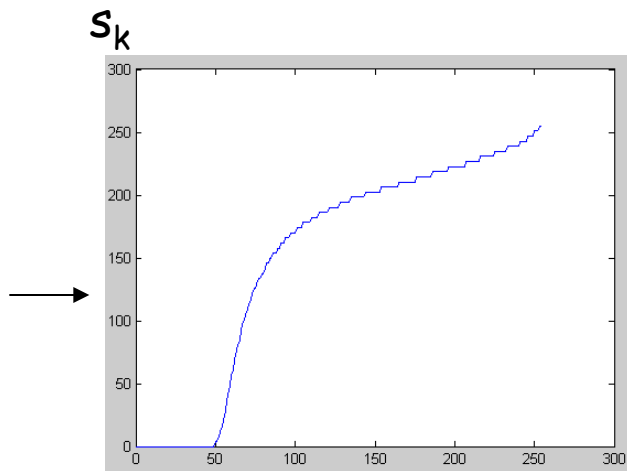
To normalize, let y axis = $p(r_k) = n_k / N$



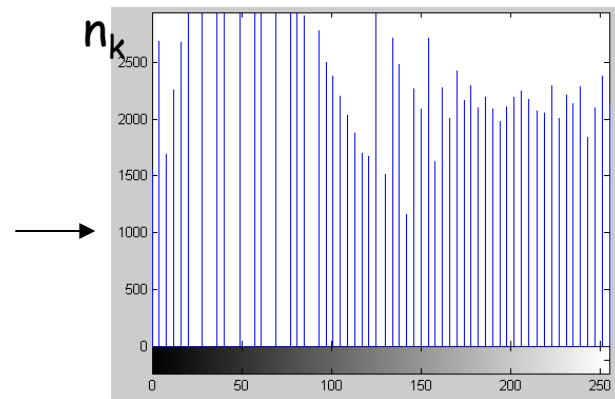
Example



r_k



$T(r)$



s_k



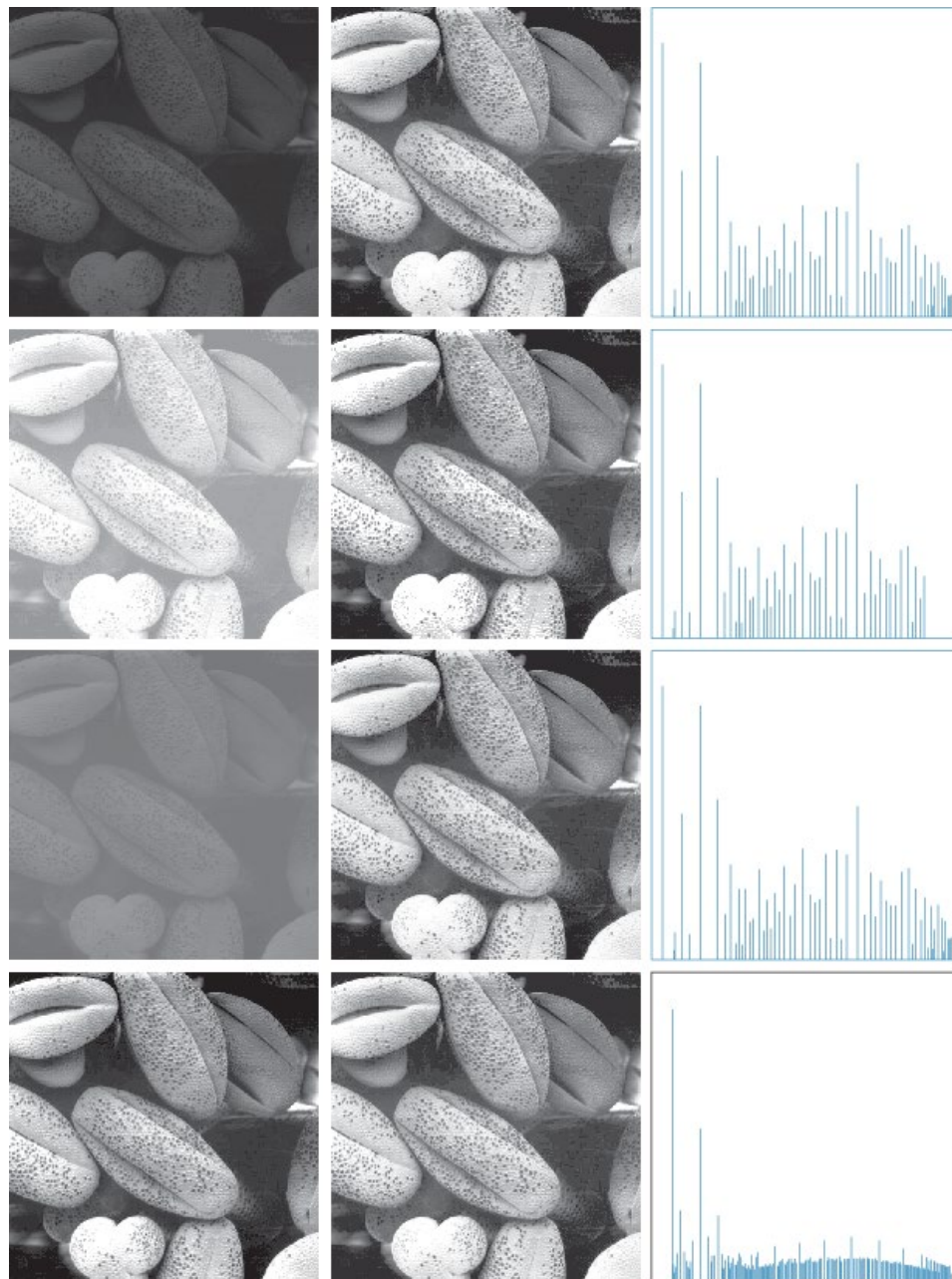


FIGURE 3.20 Left column: Images from Fig. 3.16. Center column: Corresponding histogram-equalized images. Right column: histograms of the images in the center column (compare with the histograms in Fig. 3.16).

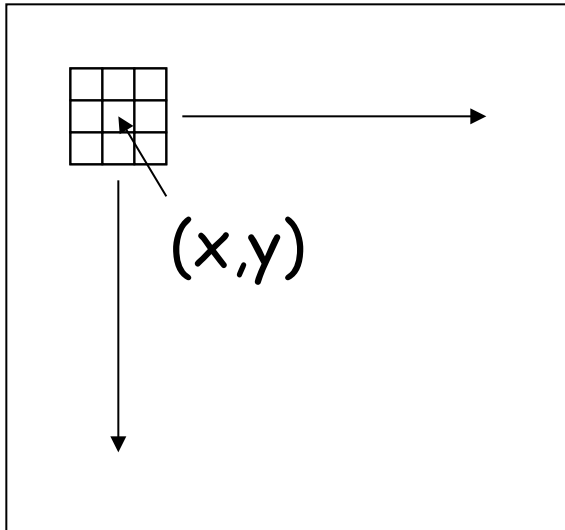
Histogram Equalization

- Can significantly improve image appearance
 - Automatic
 - users don't have to perform manual windowing or manual gray level slicing
- Nice pre-processing step before image comparison
 - Account for different lighting conditions
 - Account for different camera/device properties

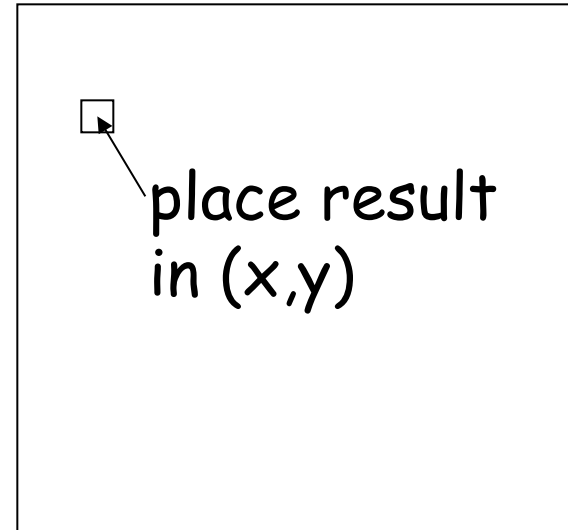
Local Enhancement

- Histogram equalization is a global operation
 - Each pixel is processed based on information from the entire image; local variations may be ignored
 - Often enhances global details
- We would like to enhance details over small areas
 - Each pixel is processed based on information from a small area/sub-image

Local Histogram Equalization



calculate histogram
using neighborhood of $m \times m$
about (x,y)



Result

Local Histogram

- Apply histogram equalization to a neighborhood around (x,y)
- Transform the gray level for pixel (x,y)
- Move the neighborhood over the rest of the image
- So, the local contrast is improved
- Example: CLAHE,
<https://pyimagesearch.com/2021/02/01/opencv-histogram-equalization-and-adaptive-histogram-equalization-clahe/>

Local Histogram

Reveals detail in local areas



Original

Global Histogram

Local Histogram

Histogram Specification

- It is useful sometimes to be able to specify the shape of the histogram that we wish the processed image to have
- Histogram specification is also known as histogram matching
- Let r be the intensity of the input image
- Let $p_r(r)$ be the PDF (probability density function) of the input image

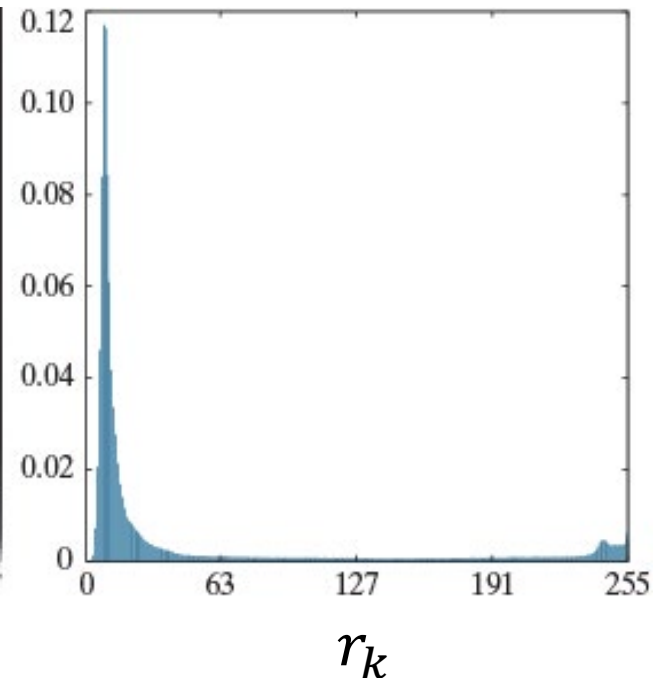
Histogram Specification

- Let $p_r(r)$ be the PDF (probability density function) of the input image

Input image



$p_r(r_k)$



a b

FIGURE 3.24

(a) An image, and
(b) its histogram.

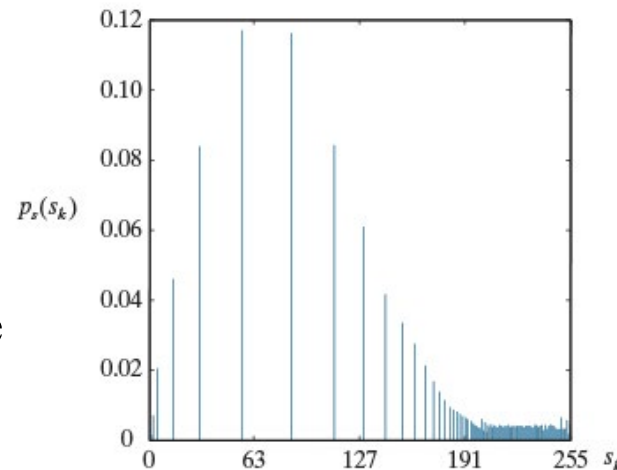
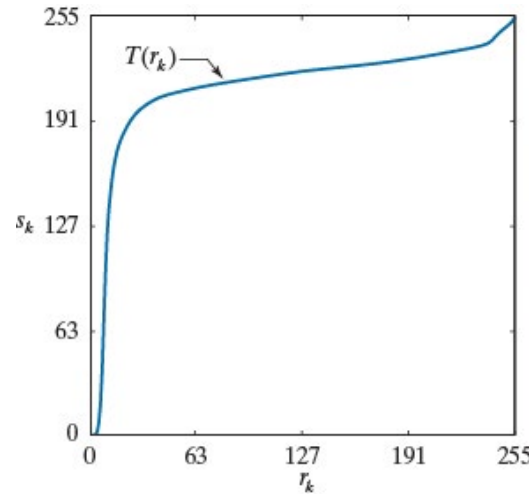
Histogram Specification

- The histogram equalized image of the input image

a b
c

FIGURE 3.25

(a) Histogram equalization transformation obtained using the histogram in Fig. 3.24(b).
(b) Histogram equalized image.
(c) Histogram of equalized image.

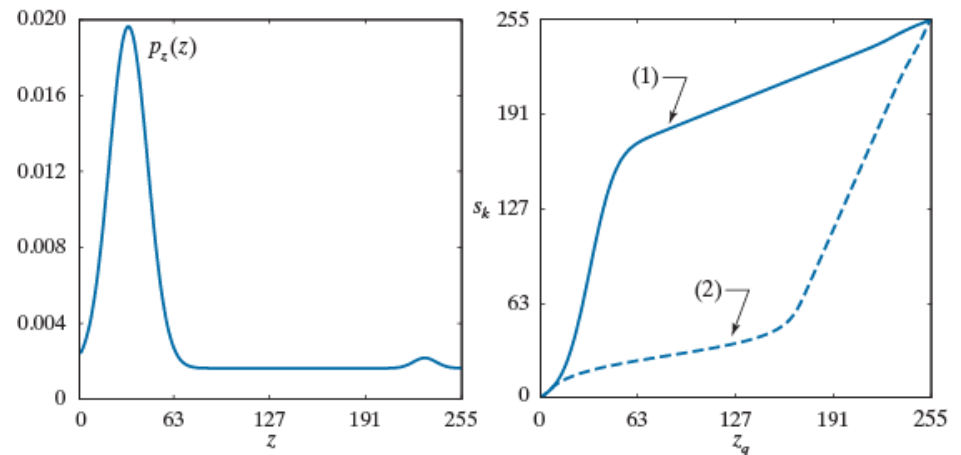


A lot of low intensity pixels are mapped to the high intensity values

Histogram Specification

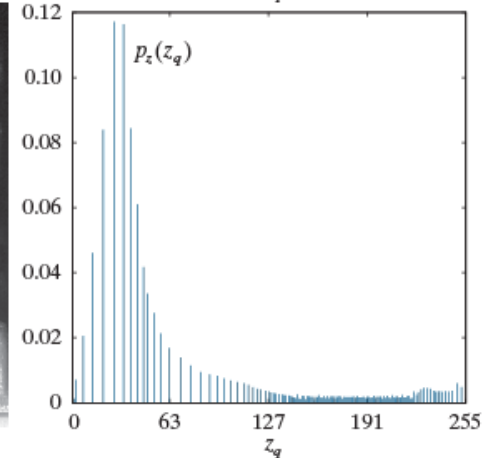
- Given the specified PDF $p_z(z)$, compute all values of the transformation function $G(z)$

$$G(Z_q) = (L - 1) \sum_{i=0}^q p_z(z_i)$$



a b
c d

FIGURE 3.26
Histogram
specification.
(a) Specified histogram.
(b) Transformation $G(z_q)$, labeled (1), and $G^{-1}(s_k)$, labeled (2).
(c) Result of histogram specification.
(d) Histogram of image (c).

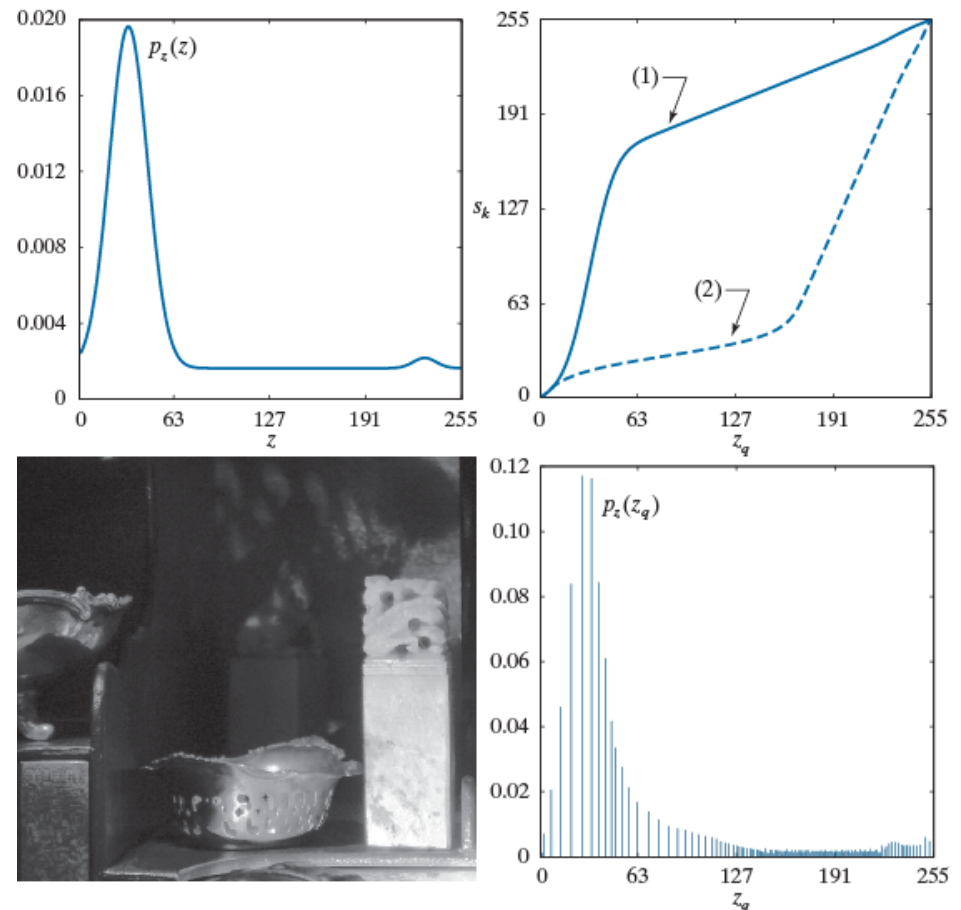


Histogram Specification

- For each value of s_k (rounded to integer), find the corresponding value of z_q so that $G(z_q)$ (rounded to integer) is closest to z_q
- Generate the histogram-specified image by mapping every equalized pixel with value s_k to the corresponding pixel with value z_q

a b
c d

FIGURE 3.26
Histogram specification.
(a) Specified histogram.
(b) Transformation $G(z_q)$, labeled (1), and $G^{-1}(s_k)$, labeled (2).
(c) Result of histogram specification.
(d) Histogram of image (c).



Histogram Specification

Source



Reference



Matched



https://scikit-image.org/docs/stable/auto_examples/color_exposure/plot_histogram_matching.html

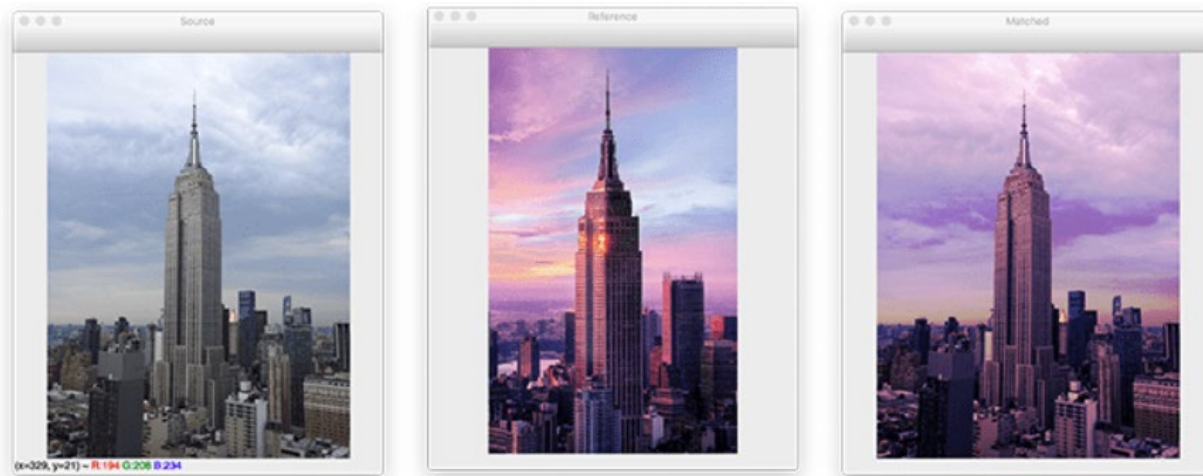
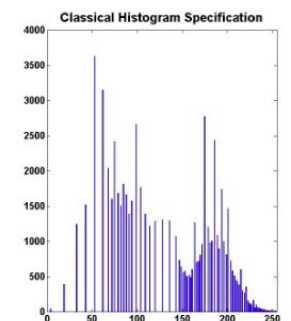
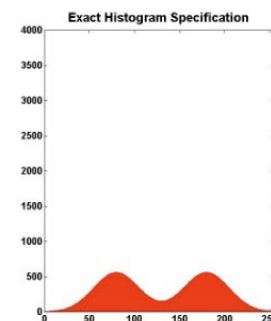
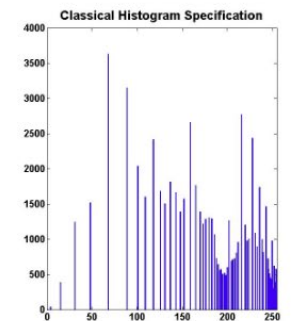
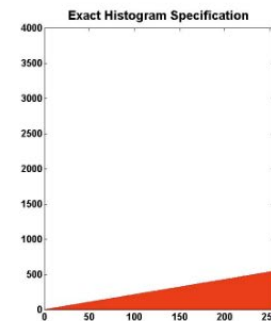
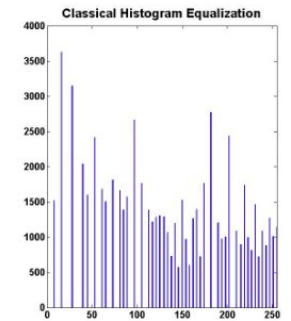
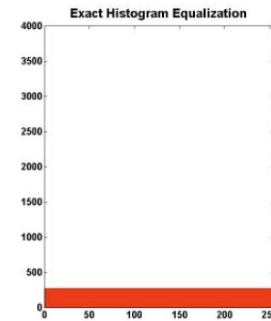
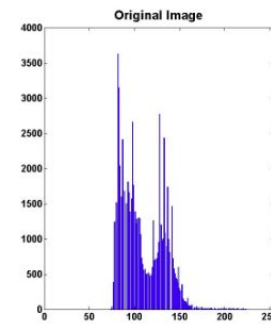


Figure 4: The output of applying histogram matching with OpenCV.

<https://pyimagesearch.com/2021/02/08/histogram-matching-with-opencv-scikit-image-and-python/>

Comparison



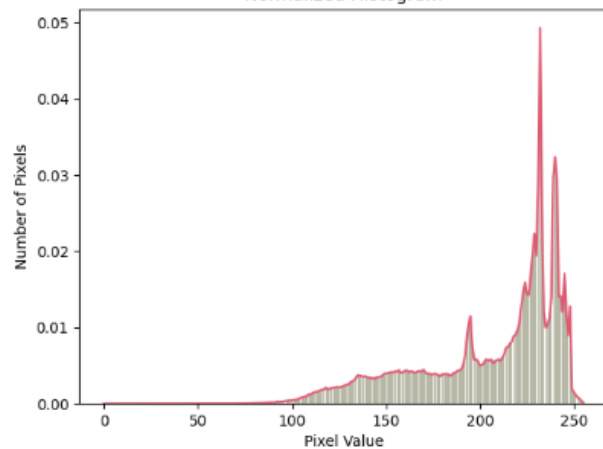
Comparison



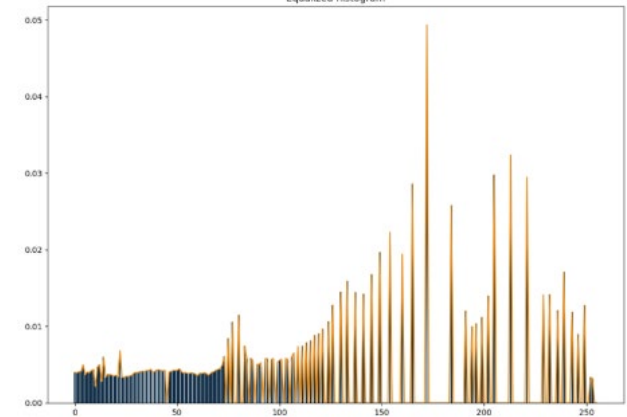
Histogram
Equalization



Normalized Histogram



Equalized histogram

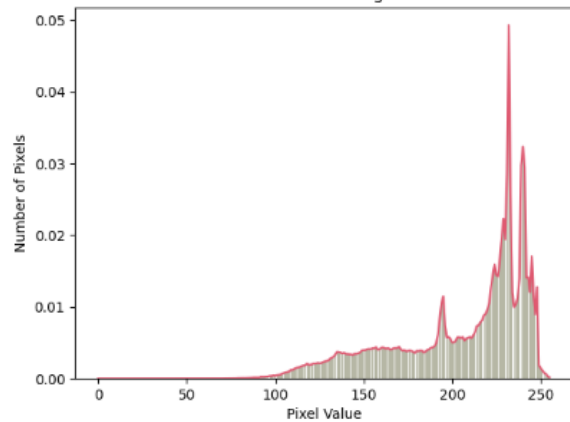


Comparison



Figure 7: Histogram matching example. We modified the left image's histogram to match the center image's histogram.

Normalized Histogram



Normalized Histogram

